

TECHSPEC® 16mm C Series Fixed Focal Length Lens

#59-870



16mm Focal Length, #59-870

TECHSPEC® C Series Fixed Focal Length Lenses are designed for use in machine vision applications with the working distance and resolution requirements of factory automation and inspection in mind. TECHSPEC® C Series Fixed Focal Length Lenses feature large maximum apertures, allowing these high performance lenses to be used in even the most restrictive lighting conditions. Each lens has a broadband anti-reflection coating, locking iris and focus adjustment with recessed set screws, and a ruggedized, robust housing for space-restrictive applications. C Series lenses are manufactured to high tolerances, which result in high levels of imaging performance with low lens-to-lens variation. Through combining optical performance, industrial features, and compact package sizes, our TECHSPEC® C Series Fixed Focal Length Lenses are ideal for use in factory automation and inspection.

General

Product Family:	C Series
Type:	Fixed Focal Length Lens
Imaging Lens Type:	Compact Lens

Physical & Mechanical Properties

Iris Option:	Variable
Length (mm):	40.50
Maximum Diameter (mm):	33
Outer Diameter (mm):	33
Weight (g):	74
Maximum Rear Protrusion (mm):	1

Optical Properties

Horizontal Field of View @ Max Sensor Format:	30.9°
Field of View at Max Sensor Format:	Horizontal: 31.0° Vertical: 23.4° Diagonal: 38.3°
Horizontal Field of View, 2/3" Sensor:	30.9°
Horizontal Field of View, 1/1.8" Sensor:	25.5°
Horizontal Field of View, 1/2" Sensor:	22.7°
Horizontal Field of View, 1/2.5" Sensor:	20.6°
Horizontal Field of View, 1/3" Sensor:	17.1°
Horizontal Field of View, 1/4" Sensor:	12.8°
Maximum Image Circle (mm):	11.00
Numerical Aperture NA, Object Side:	0.0402
Number of Elements (Groups):	7 (6)
Focal Length FL (mm):	16.00
Working Distance (mm):	100 - ∞
Aperture (f/#):	f/1.6 - f/16
Coating:	425 - 675nm BBAR
Coating Specification:	425 - 675nm BBAR
Entrance Pupil Position (mm):	24.42
Object Space Principal Plane (mm):	21.31
Image Space Principal Plane (mm):	-3.00
Maximum Distortion (%):	-0.77
Exit Pupil Position (mm):	-5.43

Sensor

Maximum Sensor Format:	2/3"
Pixel Size (µm):	2.74

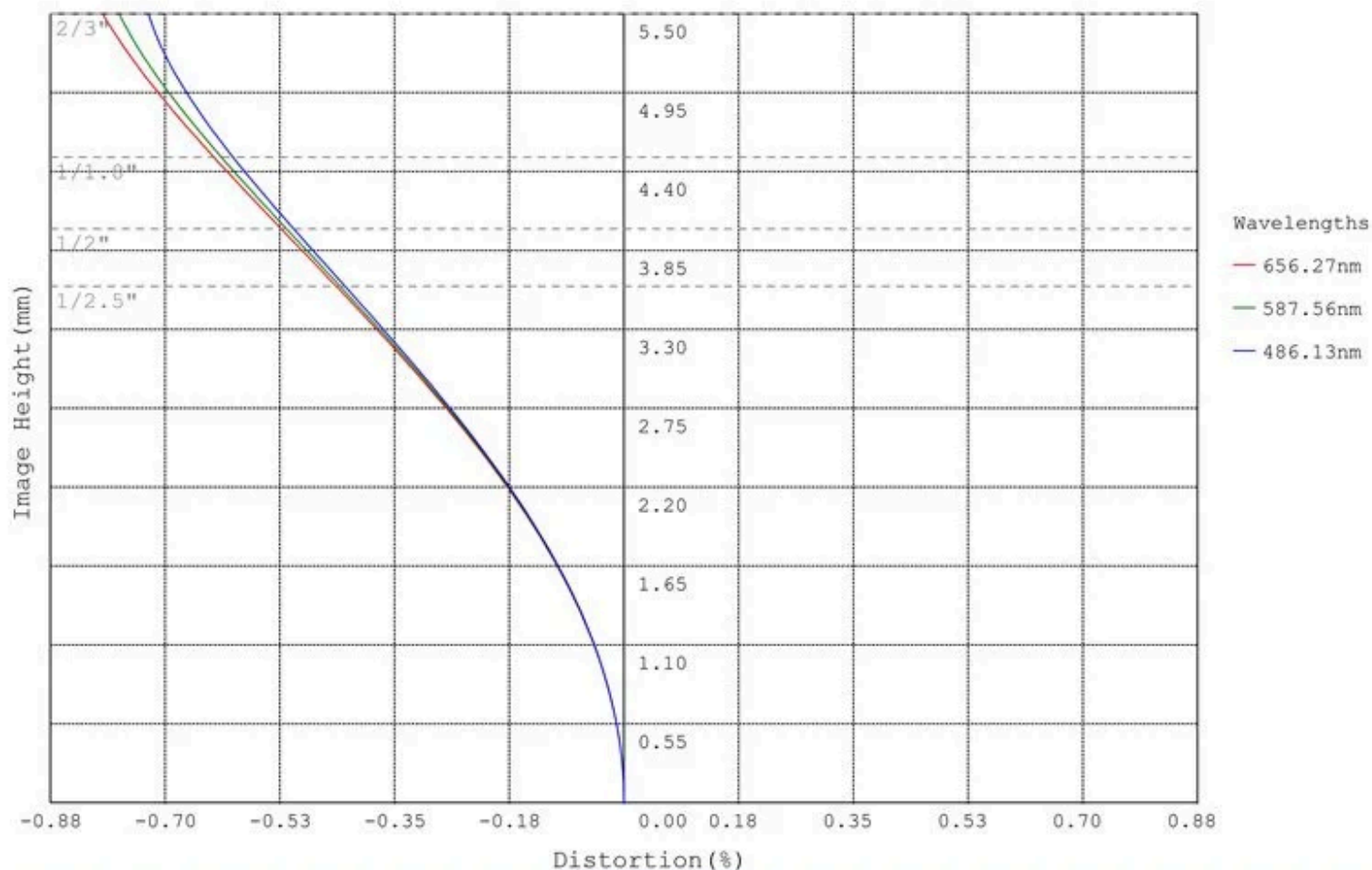
Threading & Mounting

Filter Thread:	M25.5 x 0.50 (Female)
Mount:	C-Mount

Environmental & Durability Factors

Storage Temperature (°C):	-20 to +60
---------------------------	------------

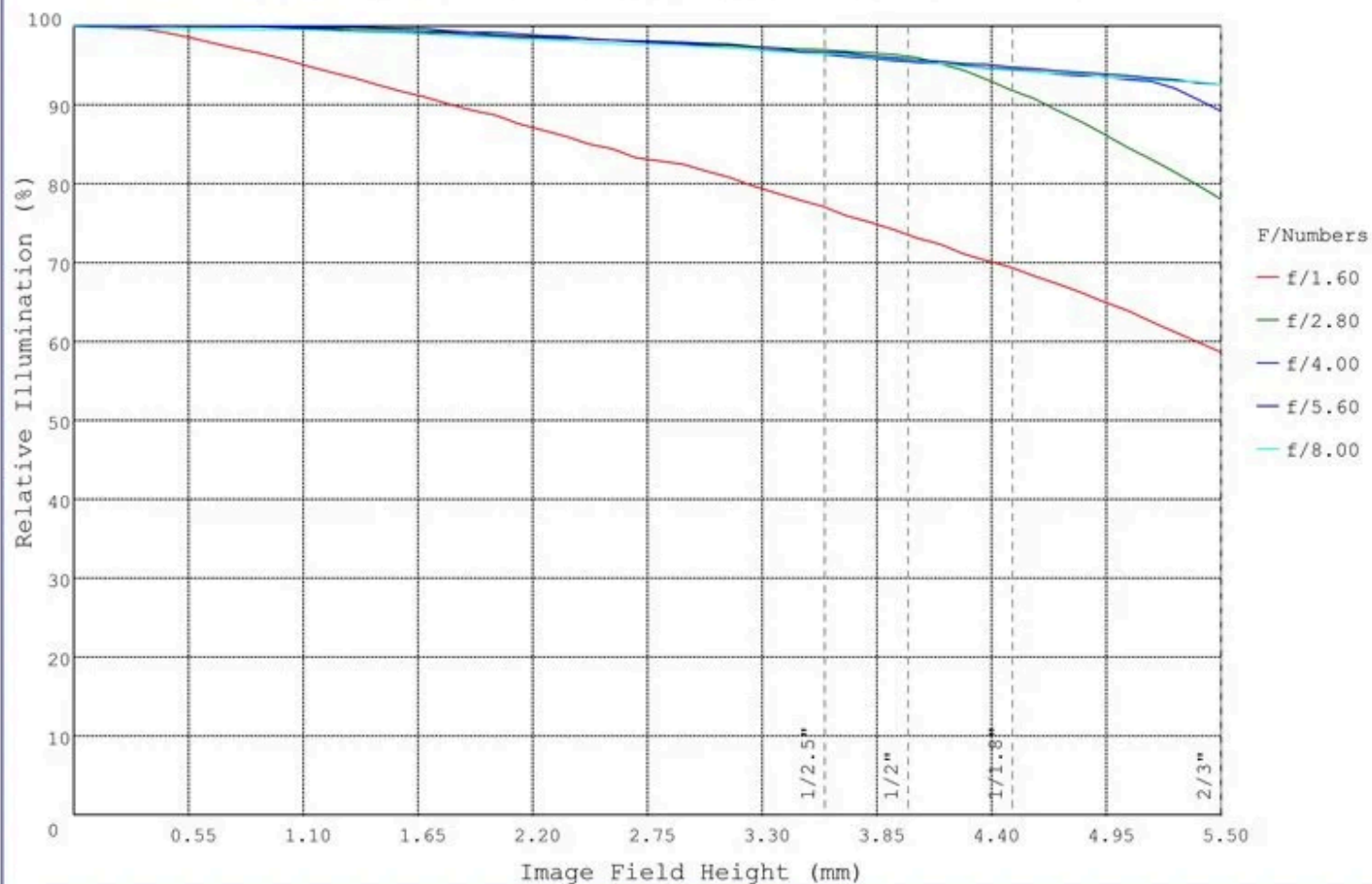
Image Height vs. Distortion: 59-870, 16mm



Working Distance: Infinity

#59-870, 16mm C Series Fixed Focal Length Lens, Distortion Plot

Relative Illumination Plot: 59-870, 16mm

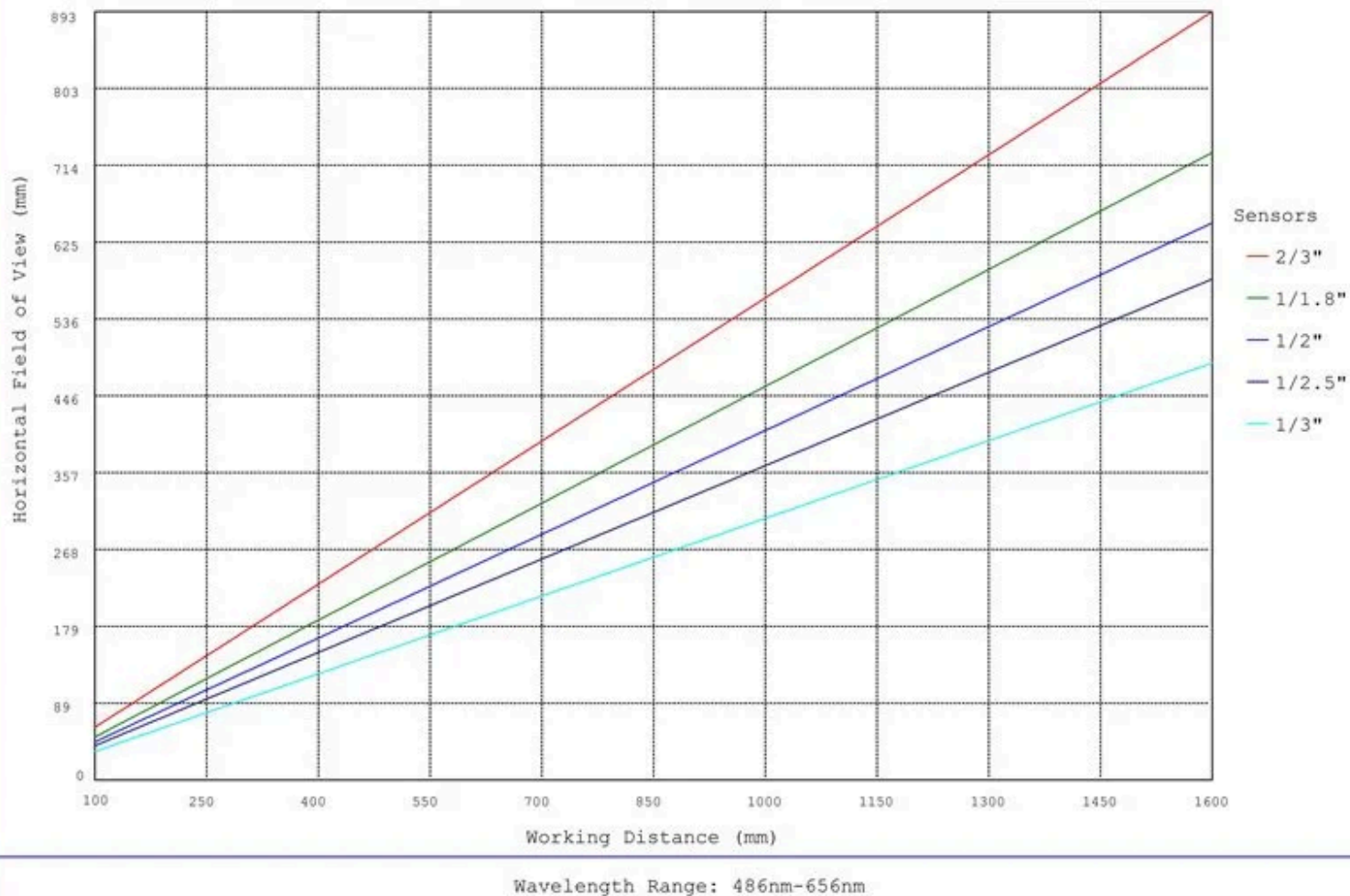


Working Distance: 100.00mm

Wavelength Range: 486nm-656nm

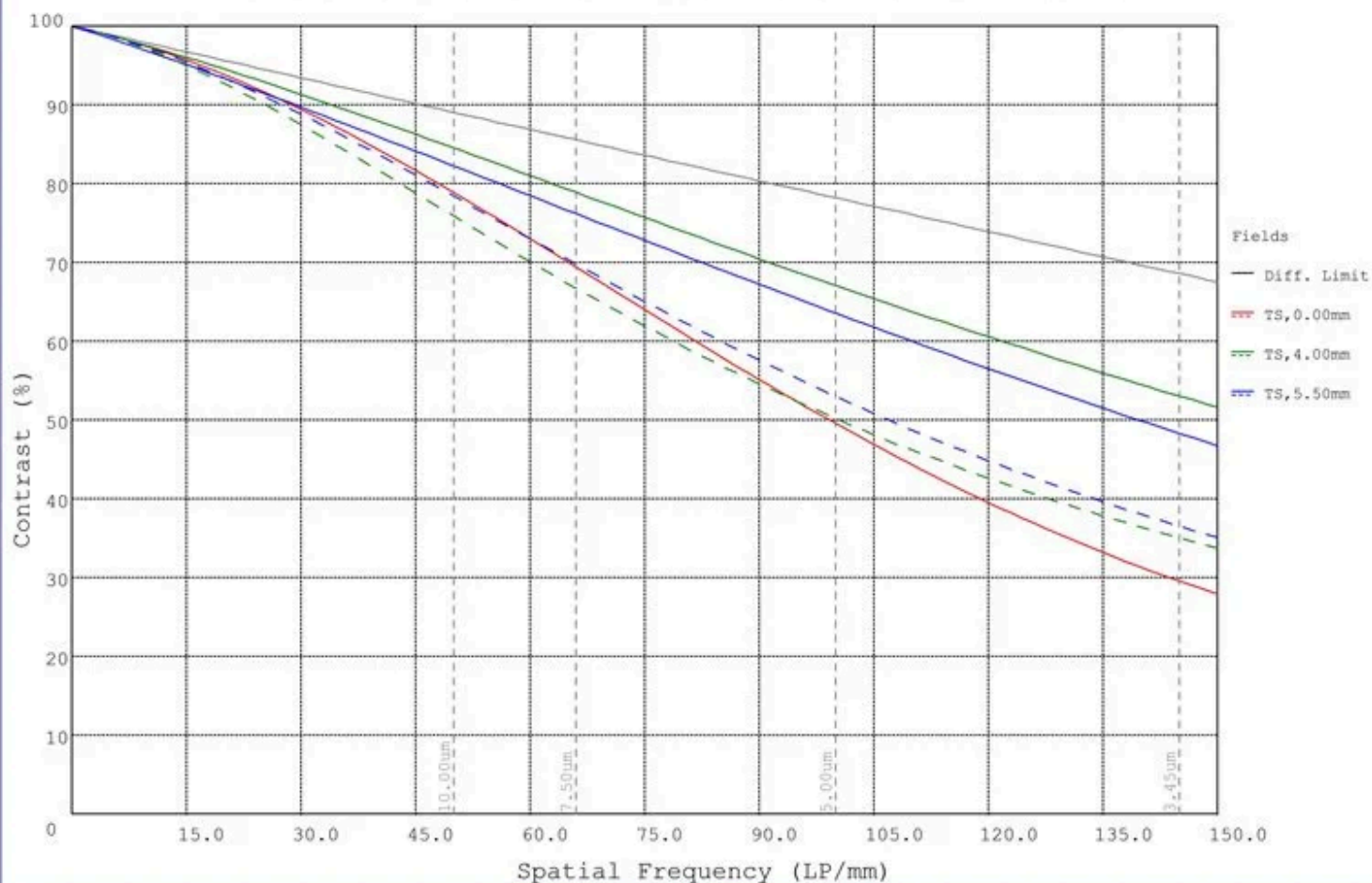
#59-870, 16mm C Series Fixed Focal Length Lens, Relative Illumination Plot

Working Distance vs. Field of View: 59-870, 16mm



#59-870, 16mm C Series Fixed Focal Length Lens, Working Distance versus Field of View Plot

Modulation Transfer Function: 59-870, 16mm



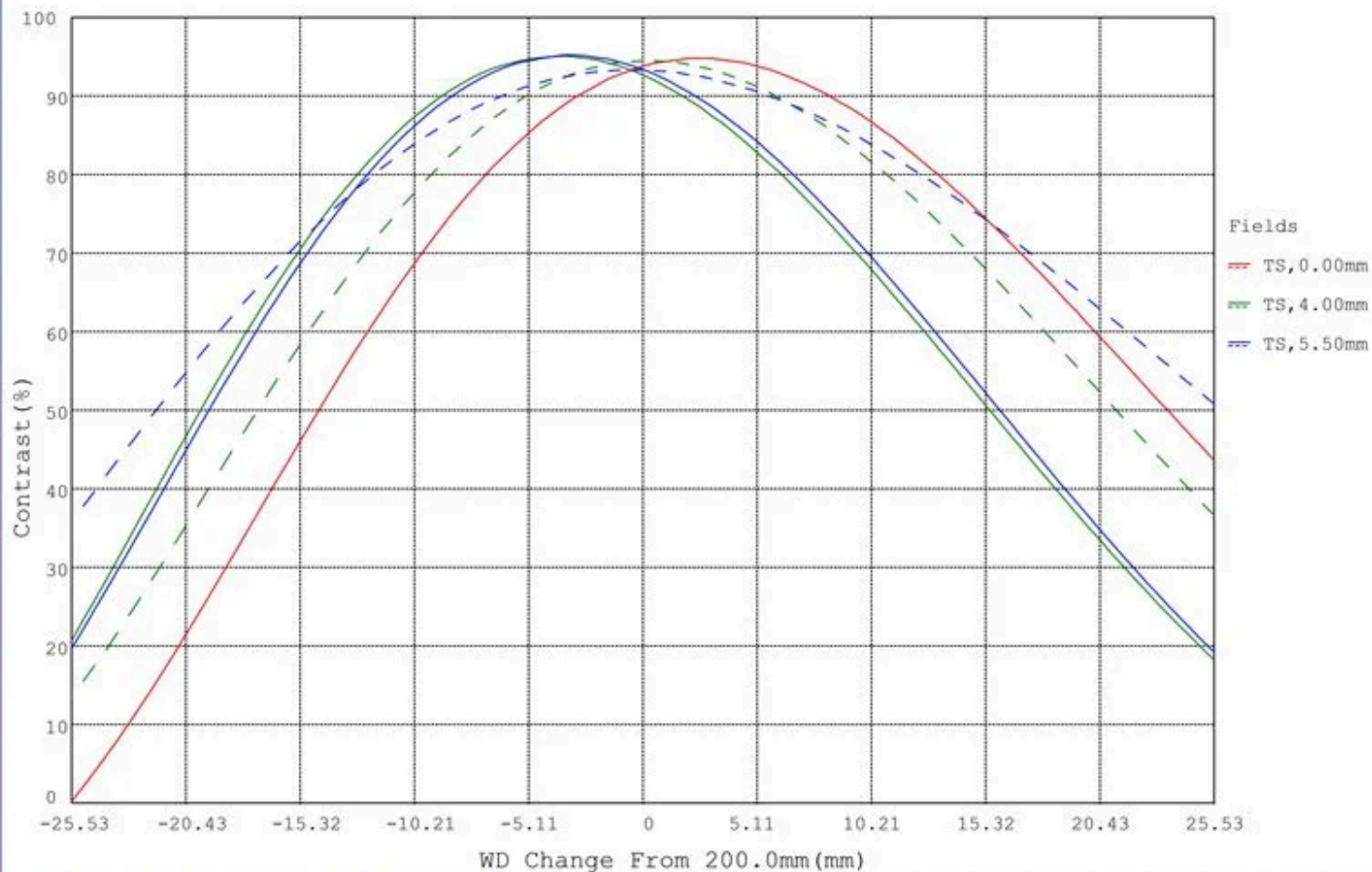
Working Distance: 200.00mm

F/Number: f/2.80

Wavelength Range: 486nm-656nm

#59-870, 16mm C Series Fixed Focal Length Lens, Modulated Transfer Function (MTF) Plot, 200mm Working Distance, f2.8

Depth of Field: 59-870, 16mm



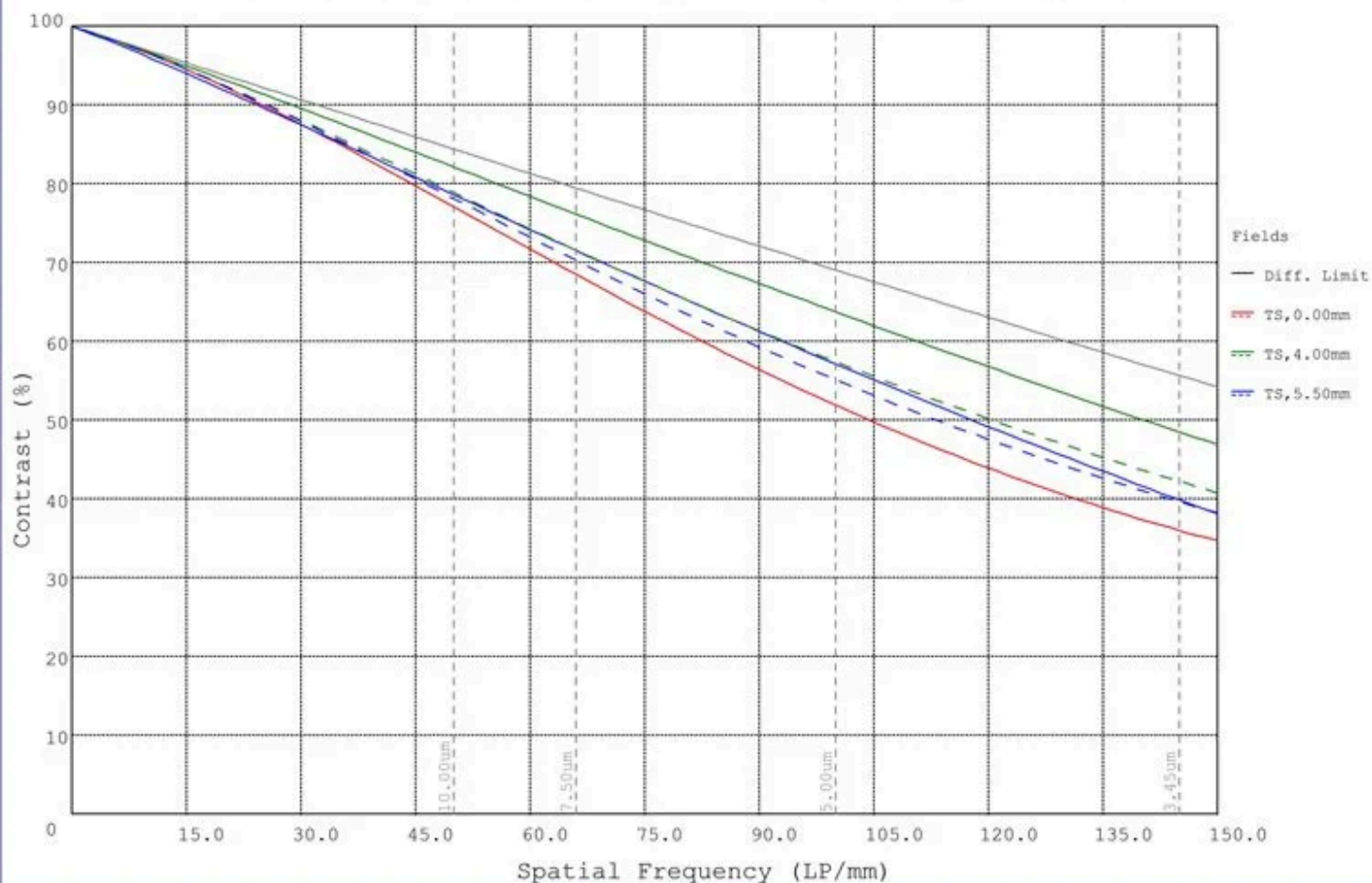
Working Distance: 200.00mm

F/Number: f/2.80

Wavelength Range: 486nm-656nm

#59-870, 16mm C Series Fixed Focal Length Lens, Depth of Field Plot, 200mm Working Distance, f2.8

Modulation Transfer Function: 59-870, 16mm



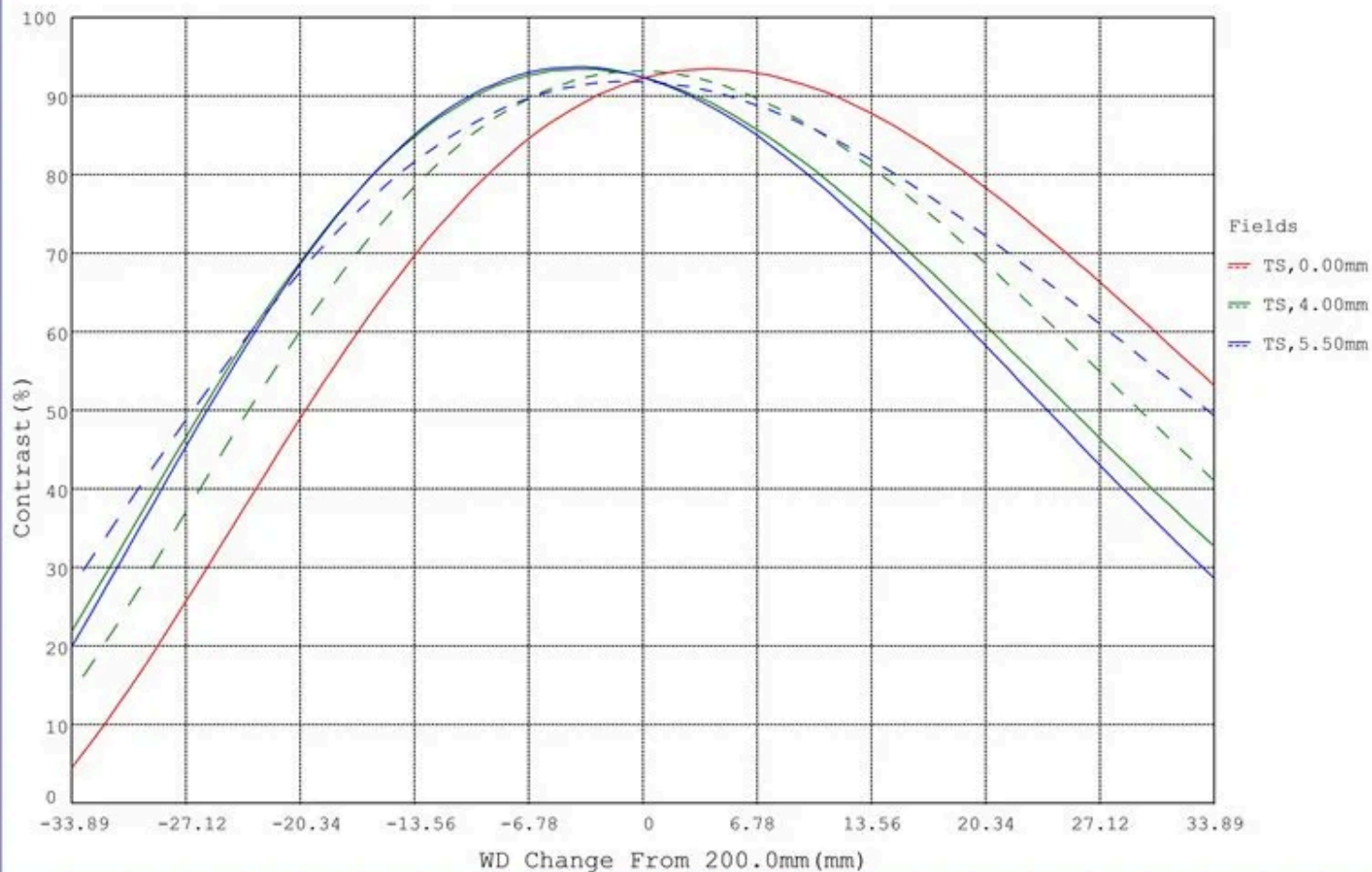
Working Distance: 200.00mm

F/Number: f/4.00

Wavelength Range: 486nm-656nm

#59-870, 16mm C Series Fixed Focal Length Lens, Modulated Transfer Function (MTF) Plot, 200mm Working Distance, f4

Depth of Field: 59-870, 16mm



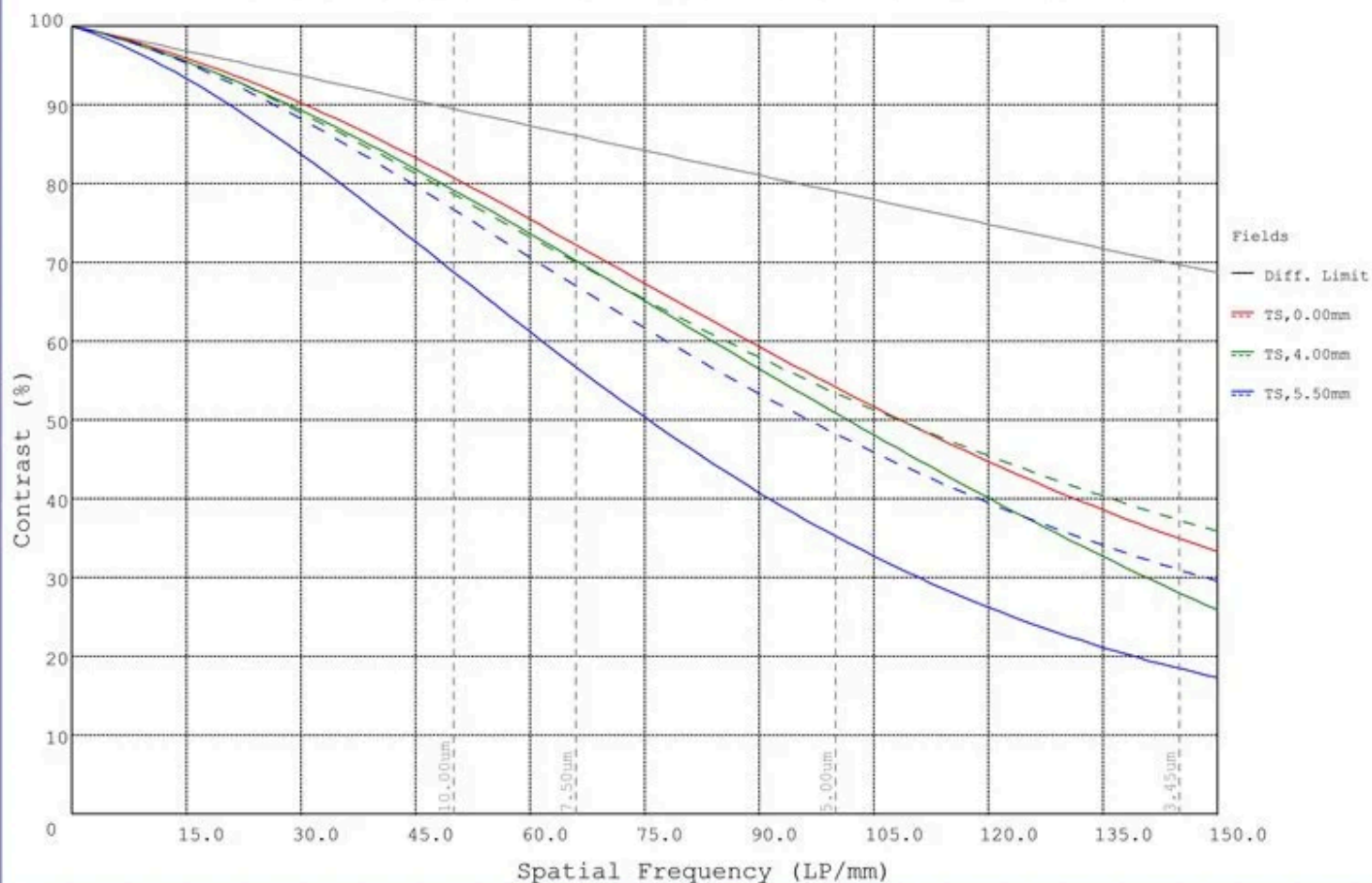
Working Distance: 200.00mm

F/Number: f/4.00

Wavelength Range: 486nm-656nm

#59-870, 16mm C Series Fixed Focal Length Lens, Depth of Field Plot, 200mm Working Distance, f4

Modulation Transfer Function: 59-870, 16mm



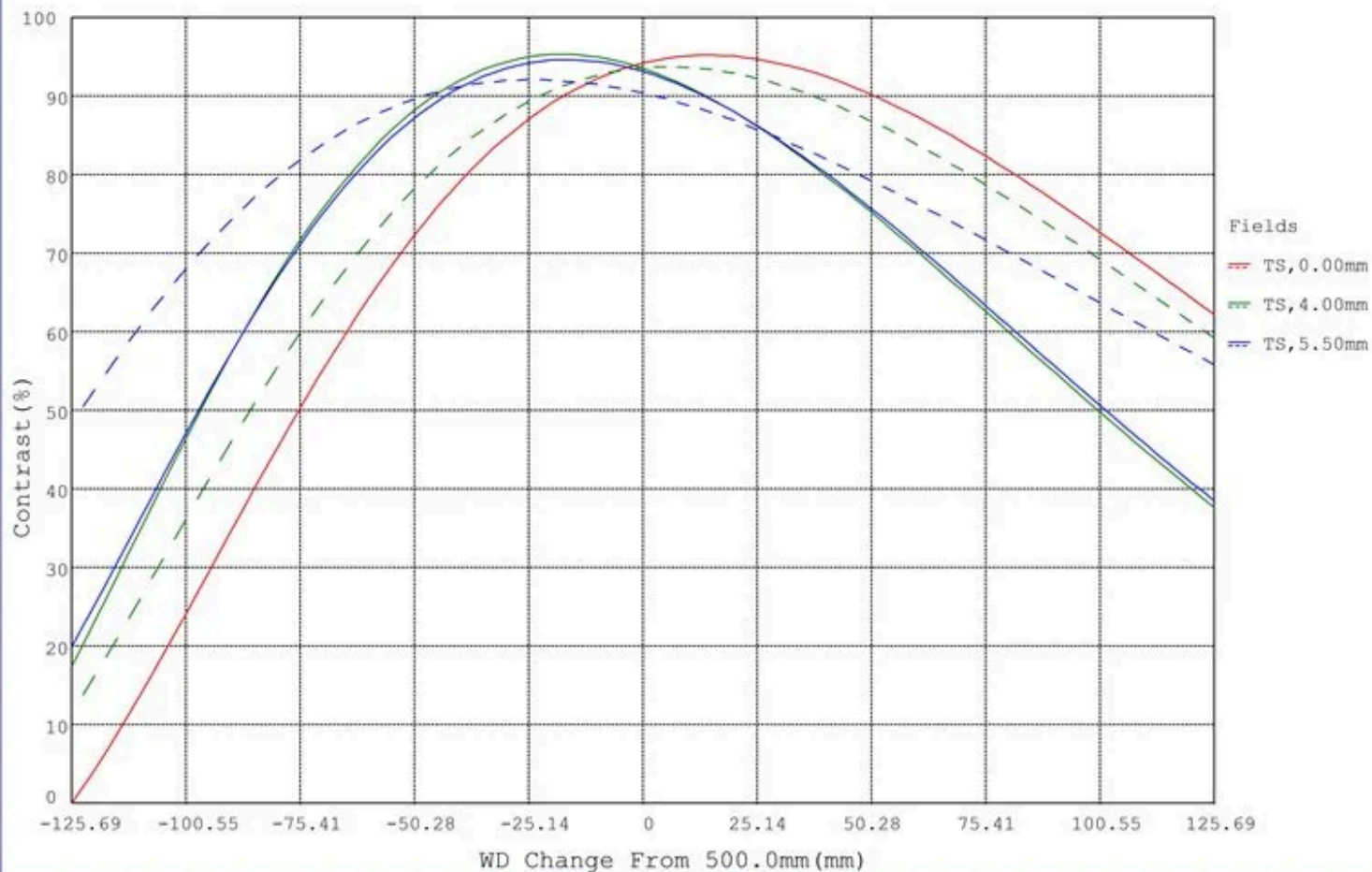
Working Distance: 500.00mm

F/Number: f/2.80

Wavelength Range: 486nm-656nm

#59-870, 16mm C Series Fixed Focal Length Lens, Modulated Transfer Function (MTF) Plot, 500mm Working Distance, f2.8

Depth of Field: 59-870, 16mm



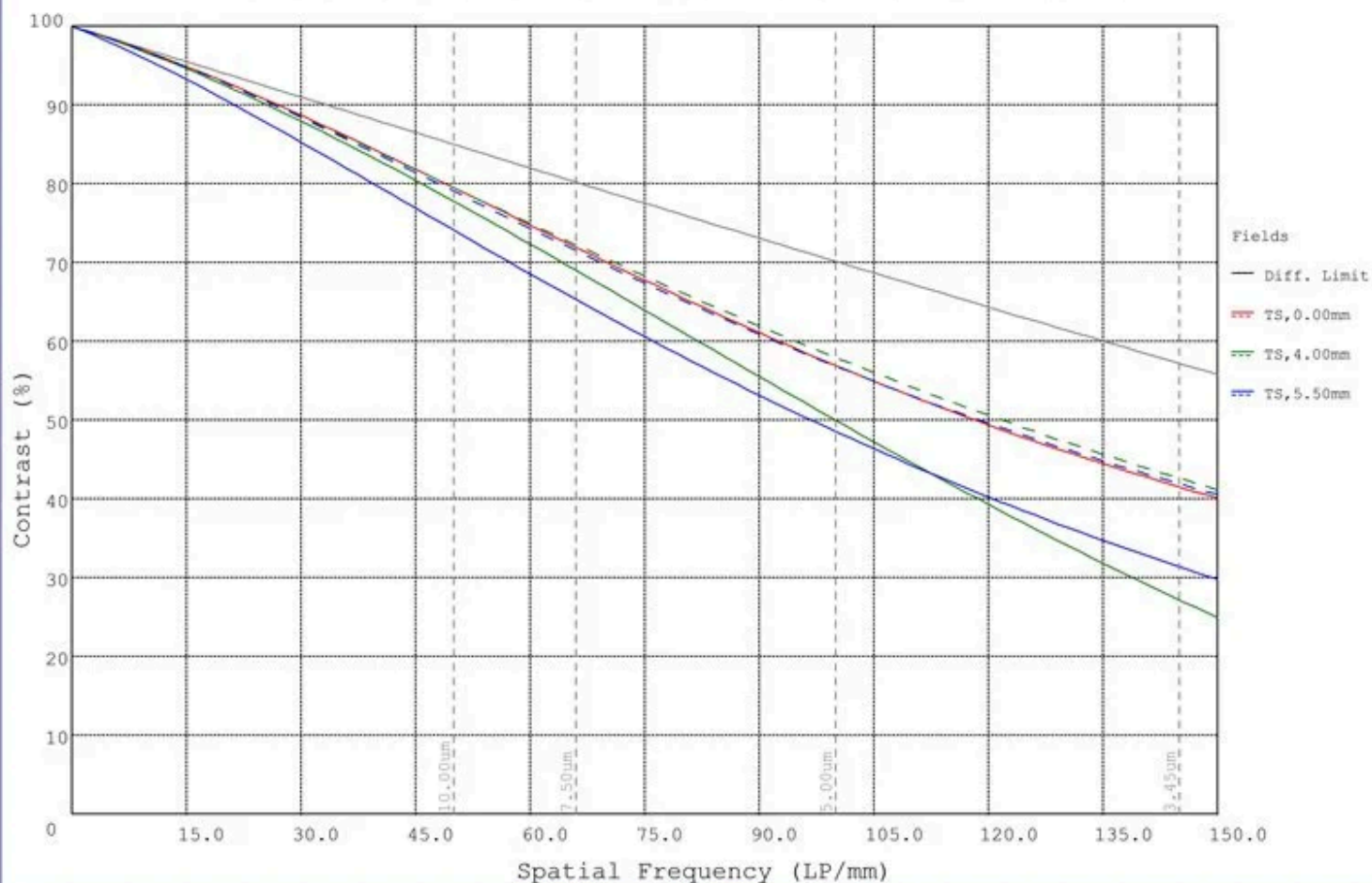
Working Distance: 500.00mm

F/Number: f/2.80

Wavelength Range: 486nm-656nm

#59-870, 16mm C Series Fixed Focal Length Lens, Depth of Field Plot, 500mm Working Distance, f2.8

Modulation Transfer Function: 59-870, 16mm



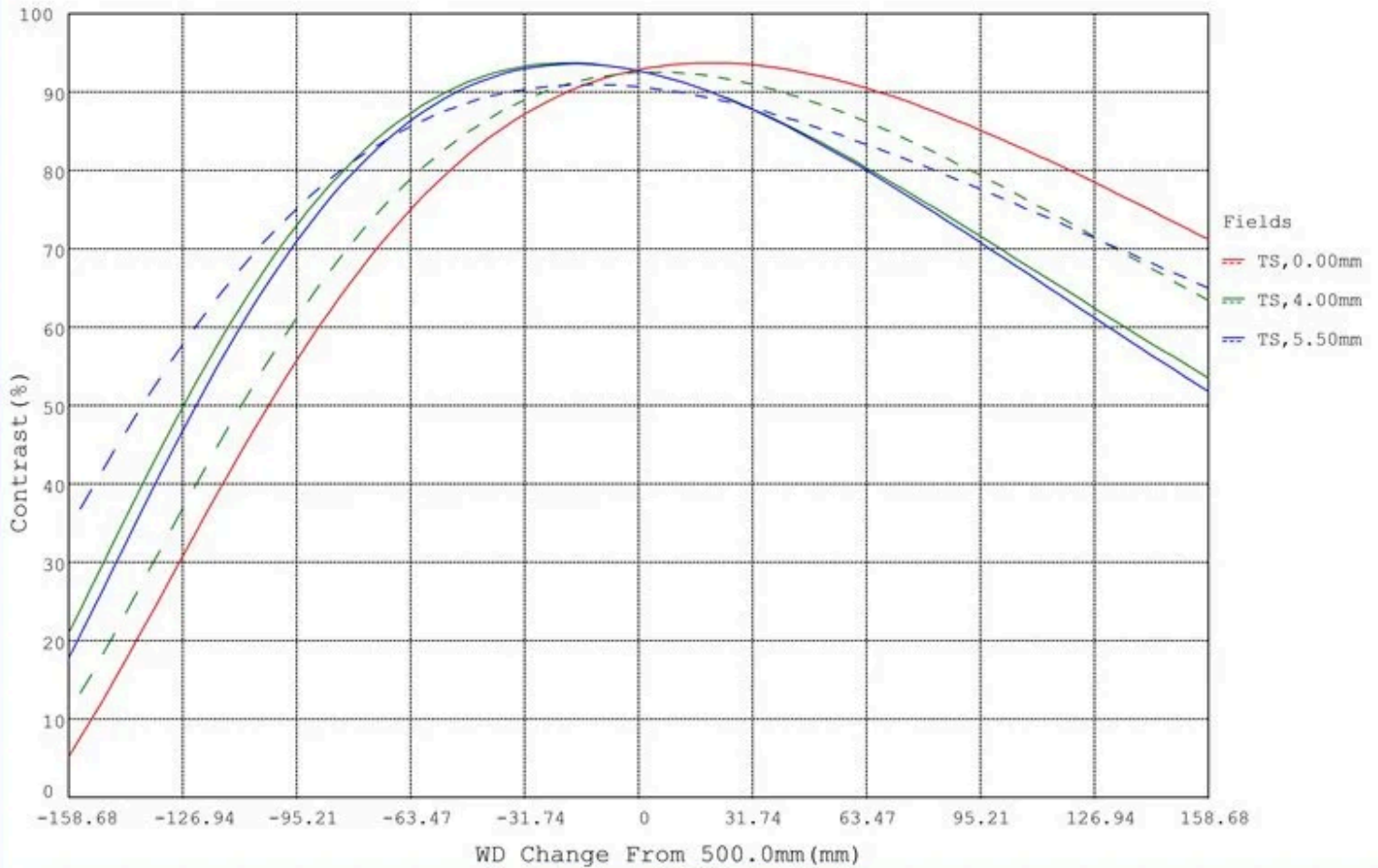
Working Distance: 500.00mm

F/Number: f/4.00

Wavelength Range: 486nm-656nm

#59-870, 16mm C Series Fixed Focal Length Lens, Modulated Transfer Function (MTF) Plot, 500mm Working Distance, f4

Depth of Field: 59-870, 16mm



Working Distance: 500.00mm

F/Number: f/4.00

Wavelength Range: 486nm-656nm

#59-870, 16mm C Series Fixed Focal Length Lens, Depth of Field Plot, 500mm Working Distance, f4



July 14, 2026

To Whom It May Concern,

This document certifies that the product stated below has been reviewed as requested by Edmund Optics:

#59-870
Subject to technical modifications (14-07-2026)



Phone: 1-800-363-1992 or 1-856-573-6250, Fax: 1-856-573-6295
101 EAST GLOUCESTER PIKE | P.O BOX 9000
BARRINGTON, NJ 08007-1380 USA
www.edmundoptics.com

Stock No.

59-870

Description

16mm C Series Fixed Focal Length Lens

This item is EU RoHS (2015/863/EU) compliant without the use of exemptions.

This certification means that:

- EO's suppliers have confirmed the material composition of this product.
- EO has implemented rigorous procedures to document this compliance.
- The information provided may, or may not, be based upon actual test data, or on information from our Vendors, Raw Material Suppliers or Subcontractors.



Jay Budd, Director of Corporate Compliance
July 14, 2026

Edmund Optics Inc. - 101 E Gloucester Pike, Barrington, NJ 08007 | 1-800-363-1992 | Compliance@edmundoptics.com



July 14, 2026

<i>Stock No.</i>	<i>Description</i>
59-870	16mm C Series Fixed Focal Length Lens

Edmund Optics certifies that all articles included in this shipment are in compliance with the terms and conditions of this order. The company also certifies that the articles included in this shipment are in accordance to all agreed upon specifications and quality assurance provisions. Please call 1-800-363-1992 should any questions arise from this shipment.



Jeff Harvey- EVP of Operations



July 14, 2026

To Whom It May Concern,

This document certifies that the product stated below has been reviewed as requested by Edmund Optics:

Stock No.	Description
59-870	16mm C Series Fixed Focal Length Lens

This item does not contain any of the substances listed on the Reach SVHC list as of 25 June 2025 in quantities exceeding 0.1% weight by weight.

Please note the following:

- Entry 11 of Annex V of the REACH regulation EC 1907/2006 as amended by regulation 987/2008, exempts “non-hazardous” glass components from registration.
- Items considered to be packing materials and which do not form part of, or are not required for the operation of, the product, are not considered to be within scope of this declaration.
- EO does not routinely analyze products for substances not purposely added or specified within the design specifications, drawings and assembly procedures.

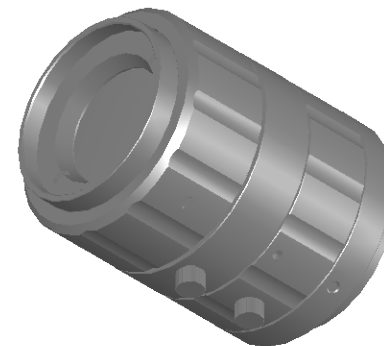
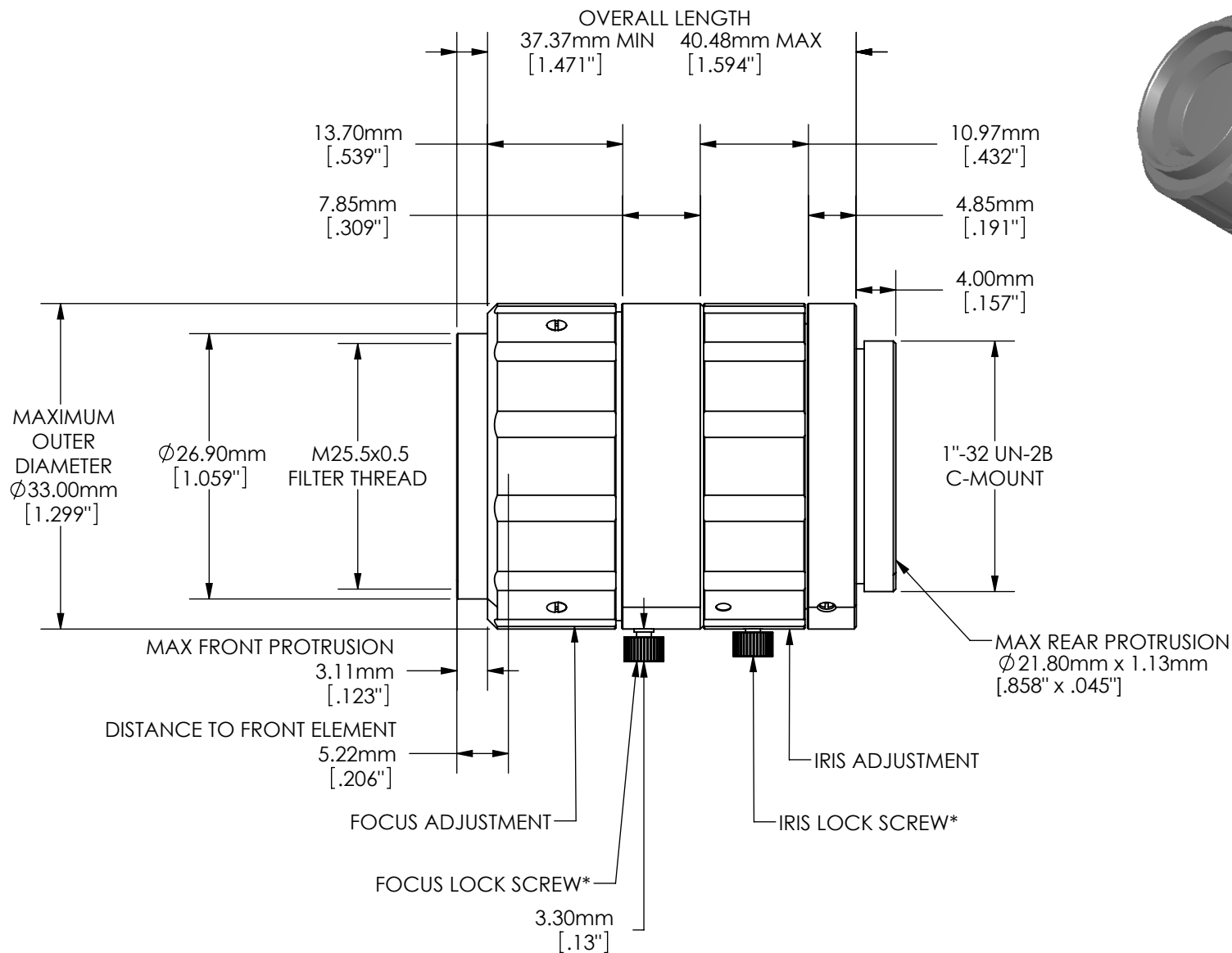
This certification means that:

- EO's suppliers have confirmed the material composition of this product.
- EO has implemented rigorous procedures to document this compliance.
- The information provided may, or may not, be based upon actual test data, or on information from our Vendors, Raw Material Suppliers or Subcontractors.



Jay Budd, Director of Corporate Compliance
July 14, 2026

Edmund Optics Inc. - 101 E Gloucester Pike, Barrington, NJ 08007 | 1-800-363-1992 | Compliance@edmundoptics.com



*ALSO INCLUDES RECESSED SET SCREW OPTION

**FOR INFORMATION ONLY:
DO NOT MANUFACTURE
PARTS TO THIS DRAWING**

SPECIFICATIONS SUBJECT TO CHANGE
WITHOUT NOTICE.
DIMENSIONS ARE FOR REFERENCE ONLY.
FOR DETAILED SPECIFICATIONS,
PLEASE REFER TO THE SPEC SHEET.



Edmund Optics

TITLE 16mm C Series Fixed Focal Length Lens

DWG NO 59870

SHEET
1 OF 1

Enable 3D View